

The system enables automatic vehicle recognition, digital payments, and contactless entry and exit. Key hardware components include RFID readers, boom barriers, and AI-enabled automatic number plate recognition (ANPR) cameras, typically equipped with integrated controllers such as Jetson Nano modules that process captured images to extract data.

Dhananjaya explains that the data supports facial recognition, license plate detection, and identifying unusual behaviour, such as fence jumping. A control box, referred to as the local processing unit (LPU), serves as the system's core.

"Most of the components used are IP-rated, ensuring they are airtight and dustproof. Cameras are designed to be fully sealed against moisture and dust. The LPU, which requires periodic maintenance and data access, is housed in a specially designed enclosure that allows for effective cooling while maintaining airtight and watertight integrity," he adds.

The system can function solely with an RFID reader or solely with a camera, or even be operated manually or remotely if neither is preferred. It is already compatible with modern connectivity options, including in-house internet, Bluetooth, Wi-Fi, and 5G. For clients concerned about data privacy, the system can also be configured to run on a local server, keeping all data on-premises rather than in the cloud.

Explaining the meaning behind the name ParkMate, Dhananjaya shares that it signifies a dependable companion that helps users find parking when they need it most.

While specific details about the R&D team structure remain confidential, Dhananjaya reveals that it includes a skilled Chief Technology Officer (CTO), a Head of Engineering



Abhimanyu Singh (left) and Dhananjaya Bharadwaj (right), Founders of ParkMate

(HOE), and a dedicated AI Engineering Manager. The team is currently placing a strong emphasis on computer vision technologies, with the long-term goal of replacing RFID readers entirely. Another key area of focus is on-street parking.

The startup claims the entire system is manufactured in India and uses no Chinese or European components. Manufacturing is handled through contracted facilities in Ghaziabad and the outskirts of Gurgaon, while the headquarters are located in Noida Sector 16.

In addition to offering hardware and software, the company provides professionally trained valets, wardens, marshals, and car park managers.

Dhananjaya explains that the payment structure varies based on the scope of services provided. Hardware is offered under a capital expenditure model, along with annual maintenance charges. Software services follow a subscription model, billed on a monthly basis through a licensing fee. Manpower services are treated as recurring operational expenses. Enterprise clients such as mall operators or smart city authorities handle all payments.

The startup is currently operating in twelve Indian cities, including Delhi NCR, Lucknow, Pune,

Hyderabad, Jhansi, and Moradabad. It has ongoing contracts with prominent names, including DLF, Phoenix, M3M, and government bodies in Uttar Pradesh and Telangana. Over the next year, ParkMate plans to focus its expansion efforts on Mumbai and Bengaluru, while consolidating its operations in the Delhi NCR region.

The team currently consists of around 350 people, including ground staff, with 36 members handling backend and managerial roles. A key challenge has been hiring skilled talent at competitive rates.

One of the company's most innovative offerings is its app-based Drop and Shop service, informally known as Dash. This valet-style system allows users to input a destination in the ParkMate app and drive to their location without worrying about parking. A valet is assigned to collect the car, which is then parked at the nearest authorised facility. The vehicle and driver are GPS-tagged, and customers receive complete visibility, including live tracking and images. If the car deviates from its intended route, the system immediately raises an alert. Vehicles are also covered by specialised liability insurance.

The full public launch of this consumer-facing service will be operational soon. "A well-known celebrity will be revealed as the brand ambassador at the time of launch, though the name is currently under wraps due to an NDA," informs Dhananjaya.

"We experienced a 300% growth in the last fiscal year and are projecting an 800% increase for the current year. Backed by a robust pipeline and confirmed contracts, we are targeting a 20-fold revenue growth over the next two years," Dhananjaya concludes. **EFY**

—Nitisha Dubey